

• An Overview of Linear Algebra

• → Linear Algebra Progression

↳ Vectors → Matrices → Subspaces

$$u \quad v \quad w \quad \left(\begin{smallmatrix} \text{linear} \\ \text{comb.} \end{smallmatrix} \right) \quad x_1 u + x_2 v + x_3 w = b$$

↳ Scalars.

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Combination of Columns

$$\text{LP } A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix} = b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$A \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

Solution

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 + b_1 \\ b_3 + b_2 + b_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}$$

↳ Have b values

Have $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ = diff. matrix resulting from Ax .

Translate b version of diff. $A = \begin{bmatrix} i & j & k \end{bmatrix} = \begin{bmatrix} b_1 & b_2 + b_1 & b_3 + b_2 + b_1 \end{bmatrix}$

$$x = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\Rightarrow Ax = b$$

$$x = A^{-1}b$$

↳ A inverse.

↳ Transform of A matrix into inverse.

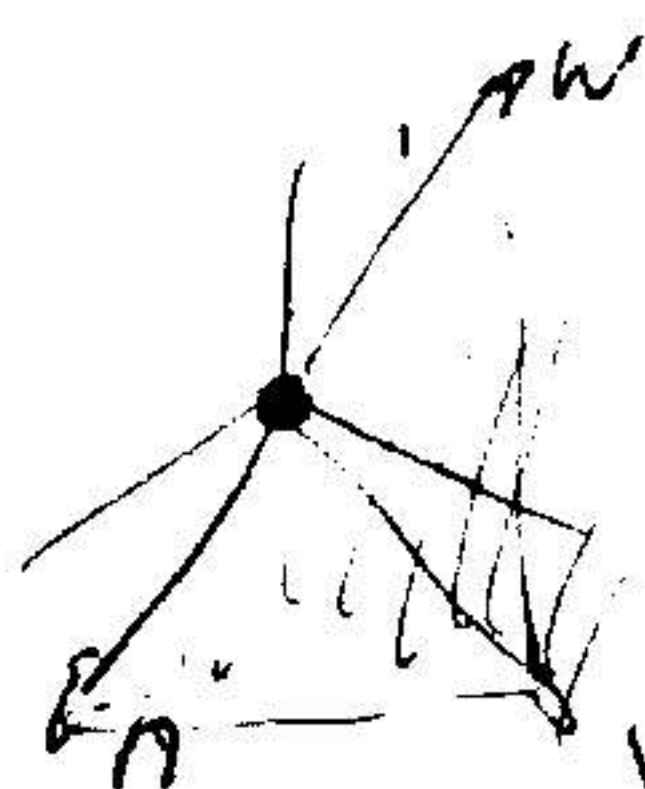
22:19/49:32

• Example 2

$$C = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Cx = 0$$

↳ Cannot get back to C using $0 < x$. 0 multiplied by x will be 0 .



→ All three in the same plane.

- Invertible Matrix $\rightarrow$ matrices for which an inverse matrix exists.

All combinations of u, v, w gives only a plane.
 $=$ all vectors Cx .
 $\rightarrow$ does not take up all the space in the cartesian plane

$$\begin{bmatrix} u & v & w \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\rightarrow$ What do we get?

$\rightarrow$ Only the bs that equal 0. $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$
 $b_1 + b_2 + b_3 = 0$.

~~$\rightarrow$ Subspace~~

$\rightarrow$ Vector Space

$\rightarrow$ A bunch of vectors where you can take all combinations of the vectors.

$\rightarrow$ Sub-Space -

$\rightarrow$ A vector space within a "larger" vector space

$\rightarrow$ Lines & Planes & Origin Point in particular are subspaces

$\rightarrow$ Rectangular Matrices

$\rightarrow$ Not invertible.

$\rightarrow$ Will rackle

$A^T A \rightarrow$ always results in square matrix

$\rightarrow$ A transpose.

A basis for $\mathbb{R}^n$ is a collection of n independent vectors in $\mathbb{R}^n$.

$\Leftrightarrow$ Equivalently, a basis is a collection of n vectors whose combinations cover the whole space.

A Vector Space is a collection of vectors that is closed under linear combinations.

A Vector Subspace is a vector space inside another vector space.

e.g. a plane through the origin in $\mathbb{R}^3$ is an example of a subspace.

Linear Algebra - Lecture 2 - 20/04/2026

Elimination with Matrices

- Elimination can succeed or fail.
 ↳ The way that all software calculates matrices.

$$\begin{array}{rclcl} x & + & 2y & + & z & = & 2 & \textcircled{1} \\ 3x & + & 8y & + & z & = & 12 & \textcircled{2} \\ & & 4y & + & z & = & 2 & \textcircled{3} \end{array}$$

$$Ax = b$$

Step 1 $\textcircled{2} - 3\textcircled{1}$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 1 \end{array} \right] \quad \begin{array}{l} \text{First pivot} \\ 1 \times 3 - \textcircled{2} \textcircled{3} \\ = 3 \times 8 - 3 \\ = 0 \end{array} \rightarrow \textcircled{2} - \textcircled{1} \textcircled{3}$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & -2 \\ 0 & 4 & 1 & 1 \end{array} \right]$$

Step 2 $\textcircled{3} - 2\textcircled{2}$
 $\textcircled{2} - 2\textcircled{1}$
 $\textcircled{2} - \textcircled{3}$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & -2 \\ 0 & 4 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & -2 \\ 0 & 0 & 5 & 5 \end{array} \right]$$

$U \rightarrow$ (assign name)

* Pivots cannot be zero

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 4 & 1 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array} \right]$$

U C

↳ Augmented matrix
 ↳ added on solution column

Back-Substitution

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array} \right) \rightarrow \begin{array}{rclcl} x & + & 2y & + & z & = & 2 & \textcircled{4} \\ & & 2y & - & 2z & = & 6 & \textcircled{5} \\ & & & & 5z & = & -10 & \textcircled{6} \end{array}$$

BS $\rightarrow$ creates new set of equations which we can use to solve the original equations with.
 ↳ Simultaneous solution.

$$\left. \begin{array}{l} z = -10 \\ y = 1 \\ x = 2 \end{array} \right\} \rightarrow \text{BS} \rightarrow \text{is the solving the equations in reverse because the system is triangular.}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

• Matrices

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \text{matrix} \times \text{column} = \text{column}$$

• Big Picture - multiplying matrices is col multiplication.

$$\begin{bmatrix} 1 & 2 & 7 \end{bmatrix} \begin{bmatrix} - \\ - \\ - \end{bmatrix} = \text{matrix} \times \text{row} = \text{row}$$

How do matrices work by row?

How do matrices work by column?

• Elimination

Step 1

• Subtract $3 \times \text{Row 1}$ from Row 2:

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

→ Elimination matrix
↳ E_{21}

Step 2: Subtract $2 \times \text{Row 2}$ from Row 3

$$E_{32} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

E_{32} Prev Version Step 2 Result.

$$\therefore E_{32}(E_{21}A_1) = U$$

$$(E_{32}E_{21})A = U$$

(Matrix Notation)

→ Cannot move multiplication order (cannot do $E_{21}E_{32}$)

→ Can reorder parenthesis
↳ which to multiply in the same order

• Permutation

Exchange Row 1 and Row 2.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

Exchange Left Column 1 and Column 2 Row Operation

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}$$

Left Right

Column Operation

- How to get from U back to A ?

$$(E_{22}E_{21})A = U$$

• Inverse Matrices

Subtract $(-3) \times$ Row 1 from Row 2

• Inverses $\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\downarrow$ inverse

Add $(3) \times$ Row 1 to Row 2

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$E^{-1} \qquad E \qquad I$

Lecture Overview

Discuss Method of Elimination

$\hookrightarrow$ A simple algorithm described in terms of matrix multiplication.

Text Book Notes = Gaussian Elimination.

Method of Elimination process $\rightarrow$ Section 2.2 page 46.

- Elimination produces an upper triangular system. $\rightarrow$ this is the goal
- Back-substitution is the process of solving the upper triangular system from the bottom upwards.

Pivot $\rightarrow$ first nonzero in the row that does the elimination.

Multiplier $\rightarrow$ (entry to eliminate) divided by (pivot) = $\frac{3}{4}$.

- To solve n equations we want to n pivots.

- Elimination Failure

$\hookrightarrow$ Method might ask us to divide by zero. $\rightarrow$ process stops

$\hookrightarrow$ Either adjust (if possible) and continue.

① e.g. $0y = 8 \rightarrow$ no solution

② $0y = 0 \rightarrow$ too many solutions.

- $\hookrightarrow$ Third means of failure (salvageable):

e.g. 1st row as zero in pivot position $\rightarrow$ "Oh no!"

Instead, we exchange it with the row below $\rightarrow$ "salvage"

Linear Algebra 1865

Lecture 3 - Multiplication & Inverse Matrices

↳ 4 ways to perform matrix multiplication.

Method 1

$$\text{Row 3} \left[\begin{array}{|c|} \hline \text{---} \\ \hline \end{array} \right] \left[\begin{array}{c|c} \text{col 4} \\ \hline b_{14} \\ b_{24} \\ b_{34} \end{array} \right] = \left[\begin{array}{c} c_{34} \end{array} \right]$$

$A \quad m \times n \quad n \times m \quad B \quad \dots \quad C = AB$

$$C_{34} = (\text{row 3 of } A) \cdot (\text{column 4 of } B)$$

$$= a_{31}b_{14} + a_{32}b_{24} + \dots = \sum_{k=1}^n a_{3k}b_{k4}$$

Method 2

$$\left[\begin{array}{|c|} \hline \text{---} \\ \hline \end{array} \right] \left[\begin{array}{|c|} \hline \text{---} \\ \hline \end{array} \right] = \left[\begin{array}{|c|} \hline \text{---} \\ \hline \end{array} \right]$$

$A_{m \times p} \quad B_{p \times n} \quad C_{m \times n}$

Rows of C are combinations of Rows of B .

Columns of C are combinations of columns A .

$$\left[\begin{array}{|c|} \hline \text{---} \\ \hline \end{array} \right] \left[\begin{array}{|c|} \hline \text{---} \\ \hline \end{array} \right] = \left[\begin{array}{|c|} \hline \text{---} \\ \hline \end{array} \right]$$

Method 3

$$\text{Col of } A \times \text{Row of } B$$

$m \times 1 \quad 1 \times p$

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix}$$

$m \times 1 \quad 1 \times p \quad m \times p$

Method 4

$$AB = \text{Sum of (Cols of } A) \times (\text{Rows of } B)$$

$$\begin{bmatrix} 2 & 7 \\ 3 & 8 \\ 4 & 9 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} + \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \begin{bmatrix} 0 & 0 \end{bmatrix}$$

Block Multiplication : Cut into blocks

$$\left[\begin{array}{c|c} A_1 & A_2 \\ \hline A_3 & A_4 \end{array} \right] \left[\begin{array}{c|c} B_1 & B_2 \\ \hline B_3 & B_4 \end{array} \right] = \left[\begin{array}{c|c} \text{---} & \text{---} \\ \hline \text{---} & \text{---} \end{array} \right]$$

$A \quad B \quad C$

Inverse Matrices

Square Matrix

↳ Can be invertible but not all are.

$$A^{-1}A = I = AA^{-1}$$

↑ If this exists

• Invertible, nonsingular

• Singular Case → No inverse matrix.

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \rightarrow \text{Why no inverse?}$$

↳ Could take determinant (= 0) to know.

↳ You can find a vector x with $Ax = 0$ (x is non zero vector)

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} x = \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Inverses

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$A \quad A^{-1} \quad I$

A = column j of A^{-1} = column j of I

Gauss-Jordan (Solve 2 eqns at once)

$$\left. \begin{array}{l} \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{array} \right\} \begin{array}{l} \begin{bmatrix} 1 & 3 & | & 1 & 0 \\ 2 & 7 & | & 0 & 1 \end{bmatrix} \xrightarrow{\text{Elimination}} \begin{bmatrix} 1 & 3 & | & 1 & 0 \\ 0 & 1 & | & -2 & 1 \end{bmatrix} \xrightarrow{} \begin{bmatrix} 1 & 0 & | & 7 & -3 \\ 0 & 1 & | & -2 & 1 \end{bmatrix} \\ \begin{matrix} A & I \end{matrix} \qquad \begin{matrix} I & A^{-1} \end{matrix} \end{array}$$

$$A^{-1}A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad \checkmark$$

$$E[A \ I] = [I \ E] = [I \ A^{-1}]$$